

Soil Health & Land Preparation

AgriMitra AI Knowledge Base — Document 6 of 6 — Soil Care Fundamentals

Common Soil Types (Andhra Pradesh Region)

Soil Type	Characteristics	Well-Suited Crops
Black cotton soil (Regur)	Deep, high clay content, good moisture retention	Cotton, sugarcane, sorghum
Red loamy soil	Well-drained, moderate fertility, common in Rayachoti	Chilli, millets, pulses
Alluvial soil (deltaic)	Fertile, fine-textured, found in Godavari/Krishna delta	Sugarcane, vegetables
Sandy/coastal soil	Light texture, low water retention, needs frequent irrigation	Coconut, groundnut

Why Soil Testing Matters

- Reveals current N-P-K status and pH, avoiding both under- and over-fertilization
- Identifies micronutrient deficiencies (Zinc, Boron, Iron) before they visibly affect the crop
- Recommended frequency: once every 2–3 years for the same field, or every season for high-value crops
- Samples should be taken from multiple spots in the field at 15–20 cm depth and mixed before testing

Soil pH and Correction

pH Range	Classification	General Correction Approach
Below 5.5	Strongly acidic	Apply agricultural lime to raise pH; improves nutrient availability
5.5 – 6.5	Slightly acidic	Generally suitable for most crops; monitor periodically
6.5 – 7.5	Neutral	Ideal range for most field crops
7.5 – 8.5	Slightly alkaline	Common in black cotton soils; organic matter addition helps
Above 8.5	Strongly alkaline	Apply gypsum; improve drainage; add organic matter over time

Improving Organic Matter & Soil Structure

- Incorporate farmyard manure (FYM) or compost before the main sowing season to build organic matter
- Practice green manuring (e.g., growing and ploughing in dhaincha or sunhemp) ahead of a nutrient-demanding crop
- Retain crop residue where feasible instead of burning it, to return organic matter and reduce air pollution
- Avoid excessive tillage, which can degrade soil structure and accelerate organic matter loss over time

Land Preparation Steps (General Sequence)

- Deep ploughing after the previous harvest to break hardpan and expose soil to sunlight (helps control soil-borne pests)
- Incorporate farmyard manure or compost during the first ploughing
- Level the field to ensure uniform water distribution during irrigation
- Prepare final tilth appropriate to the crop — fine tilth for small-seeded crops like chilli nursery beds, coarser for direct-sown crops like maize
- Form ridges/furrows or bunds as required by the crop and irrigation method

Land Preparation Notes by Crop Type

Crop	Special Land Preparation Note
Paddy	Puddling (wet ploughing) is done before transplanting to reduce water percolation and control weeds
Cotton, Maize	Deep summer ploughing helps break pest life cycles and improve water infiltration
Chilli	Raised nursery beds needed for seedling raising; main field needs fine tilth for transplanting
Groundnut	Fine tilth important since pegs must penetrate soil easily for pod formation
Sugarcane	Deep ploughing and furrow formation for sett planting; field leveling critical for irrigation uniformity

Note for verification: Soil type boundaries and correction quantities (e.g., lime or gypsum dose per acre) vary by exact soil test results. Use this document for general orientation and always refer to the specific soil test report and local KVK/agriculture officer for quantities.